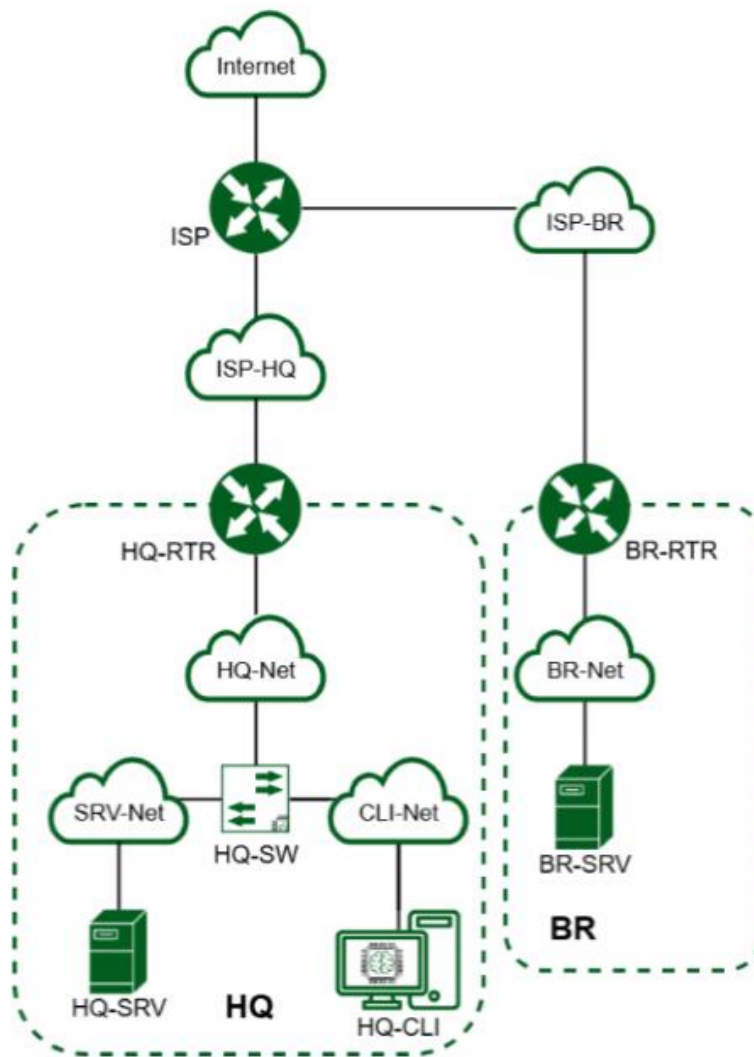


Топология сети:



Используемые ОС:

ISP, HQ-RTR, BR-RTR - Astra Linux CE

HQ-SRV, BR-SRV - Альт Сервер 10

HQ-CLI - Альт Рабочая станция 10

Таблица адресации:

Устройство	Интерфейс	IP адрес	Маска подсети	Шлюз по умолчанию
ISP	eth0	DHCP (автоматически)	-	DHCP (автоматически)
ISP	eth1	172.16.1.1/28	255.255.255.240	-
ISP	eth2	172.16.2.1/28	255.255.255.240	-
HQ-RTR	eth0	172.16.1.2/28	255.255.255.240	172.16.1.1
HQ-RTR	eth1.100	192.168.1.1/27	255.255.255.224	-
HQ-RTR	eth1.200	192.168.2.1/28	255.255.255.240	-
HQ-RTR	eth1.999	192.168.99.1/29	255.255.255.248	-
HQ-RTR	gre1 (ip tunnel)	10.0.0.1/30	255.255.255.252	-
HQ-SRV	ens18	192.168.1.2/27	255.255.255.224	192.168.1.1
HQ-CLI	ens18	DHCP (192.168.2.2/28)	255.255.255.240	DHCP (192.168.2.1)
BR-RTR	eth0	172.16.2.2/28	255.255.255.240	172.16.2.1
BR-RTR	eth1	192.168.3.1/28	255.255.255.240	-
BR-SRV	ens18	192.168.3.2/28	255.255.255.240	192.168.3.1

Модуль 1. Настройка сетевой инфраструктуры

ISP (

Login: user

Password: P@ssw0rd):

Sudo su

hostnamectl set-hostname isp.au-team.irpo

bash

mcedit /etc/network/interfaces

```
/etc/network/interfaces [-M--] 0 L:[ 1+20 21/ 21] *(387 / 387b) <EOF>
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp

auto eth1
iface eth1 inet static
<----->address 172.16.1.1/28

auto eth2
iface eth2 inet static
<----->address 172.16.2.1/28
```

echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf

sysctl -p

iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE

iptables-save > /etc/iptables.rules

crontab -e

В КОНЕЦ ДОБАВИТЬ:

@reboot /sbin/iptables-restore /etc/iptables.rules

@reboot /sbin/sysctl -p

echo "nameserver 8.8.8.8" > /etc/resolv.conf

systemctl restart networking

```
timedatectl set-timezone Asia/Yekaterinburg
```

HQ-RTR (

Login: user

Password: P@ssw0rd):

Sudo su

hostnamectl set-hostname hq-rtr.au-team.irpo

bash

mcedit /etc/network/interfaces

```
GNU nano 2.7.4                                     файл: /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 172.16.1.2/28
    gateway 172.16.1.1

auto eth1
iface eth1 inet manual

auto eth1.100
iface eth1.100 inet static
    address 192.168.1.1/27
    vlan-raw-device eth1

auto eth1.200
iface eth1.200 inet static
    address 192.168.2.1/28
    vlan-raw-device eth1

auto eth1.999
iface eth1.999 inet static
    address 192.168.99.1/29
    vlan-raw-device eth1

auto gre1
iface gre1 inet tunnel
    address 10.0.0.1
    netmask 255.255.255.252
    mode gre
    local 172.16.1.2
    endpoint 172.16.2.2
    ttl 255
```

systemctl restart networking

echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf

sysctl -p

```
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables-save > /etc/iptables.rules
```

```
echo "nameserver 8.8.8.8" > /etc/resolv.conf
```

```
useradd net_admin -u 2026 -m -s /bin/bash
```

```
passwd net_admin
```

```
echo "net_admin ALL=(ALL) NOPASSWD:ALL" > /etc/sudoers.d/net_admin
```

```
timedatectl set-timezone Asia/Yekaterinburg
```

FRR + OSPF

```
echo "deb [trusted=yes] https://archive.debian.org/debian buster main" >>
/etc/apt/sources.list
apt-get update && apt-get install frr ssh -y
```

```
mcedit /etc/frr/daemons → ospfd=no→ospfd=yes
systemctl restart frr
```

```
vysh
```

```
conf t
```

```
router ospf
```

```
network 10.0.0.0/30 area 0
```

```
network 192.168.1.0/27 area 0
```

```
network 192.168.2.0/28 area 0
```

```
network 192.168.99.0/29 area 0
```

```
passive-interface eth0
```

```
passive-interface eth1
```

```
exit
```

```
int gre1
```

```
ip ospf authentication message-digest
```

```
ip ospf message-digest-key 1 md5 P@ssw0rd
```

end

wr mem

exit

systemctl restart frr

systemctl enable --now frr

crontab -e

в конец добавить:

@reboot /sbin/iptables-restore /etc/iptables.rules

@reboot /sbin/sysctl -p

DHCP

apt-get install -y isc-dhcp-server

mcedit /etc/default/isc-dhcp-server

```
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="eth1.200"
#INTERFACESv6=""
```

mcedit /etc/dhcp/dhcpd.conf

```
GNU nano 2.7.4                               Файл: /etc/dhcp/dhcpd.conf
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#
# option definitions common to all supported networks
#option domain-name "example.org";
#option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 600;
max-lease-time 7200;

# The ddns-updates-style parameter controls whether or not the server will
# attempt to do a DNS update when a lease is confirmed. We default to the
# behavior of the version 2 packages ('none', since DHCP v2 didn't
# have support for DDNS.)
ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;
```

А также в конце пишем:

```
# A slightly different configuration for an internal subnet.
subnet 192.168.2.0 netmask 255.255.255.240 {
    range 192.168.2.2 192.168.2.10;
    option domain-name-servers 192.168.1.2;

    option routers 192.168.2.1;
}
```

Когда мы изменили файл для этого вида, идём дальше.

`systemctl restart isc-dhcp-server`

BR-RTR: (

Login: user

Password: P@ssw0rd):

Sudo su

hostnamectl set-hostname br-rtr.au-team.irpo

mcedit /etc/network/interfaces

mcedit /etc/network/interfaces

```
/etc/network/interfaces [----] 21 L:[ 1+24 25/ 27] *(522 / 537b) 0046 0x02E
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
    address 172.16.2.2/28
    gateway 172.16.2.1

auto eth1
iface eth1 inet static
    address 192.168.3.1/28

auto gre1
iface gre1 inet tunnel
    address 10.0.0.2
    netmask 255.255.255.252
    mode gre
    local 172.16.2.2
    endpoint 172.16.1.2
    ttl 255
```

systemctl restart networking

timedatectl set-timezone Asia/Yekaterinburg

```
echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf
sysctl -p
```

```
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables-save > /etc/iptables.rules
```

```
echo "nameserver 8.8.8.8" > /etc/resolv.conf
```

```
useradd net_admin -u 2026 -m -s /bin/bash
```

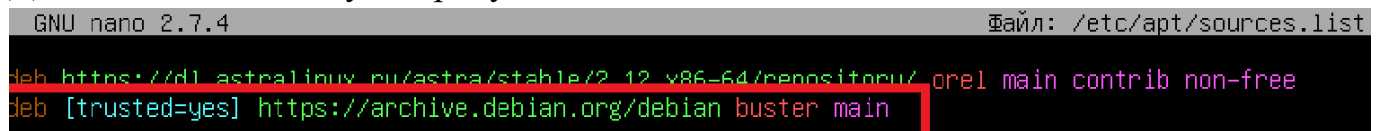
```
passwd net_admin
```

```
echo "net_admin ALL=(ALL) NOPASSWD:ALL" > /etc/sudoers.d/net_admin
```

FRR + OSPF

```
mcedit /etc/apt/sources.list
```

Добавляем выделенную строку:



```
GNU nano 2.7.4                                Файл: /etc/apt/sources.list
deb https://dl.astralinux.ru/astra/stable/2.12/x86-64/repositories/ gres1 main contrib non-free
deb [trusted=yes] https://archive.debian.org/debian buster main
```

```
apt-get update && apt-get install frr ssh
```

```
mcedit /etc/frr/daemons → ospfd=no→ospfd=yes
```

```
systemctl restart frr
```

```
vttysh
```

```
conf t
```

```
router ospf
```

```
network 10.0.0.0/30 area 0
```

```
network 192.168.3.0/28 area 0
```

```
passive-interface eth0
```

```
passive-interface eth1
```

```
exit
```

```
int gres1
```

```
ip ospf authentication message-digest
```

```
ip ospf message-digest-key 1 md5 P@ssw0rd
```

```
end
```

```
wr mem
```

```
exit
```

```
crontab -e
```

В КОНЕЦ ДОБАВИТЬ:

```
@reboot /sbin/iptables-restore /etc/iptables.rules
```

```
@reboot /sbin/sysctl -p
```

HQ-SRV(

Login: root

Password: P@ssw0rd

):

```
hostnamectl set-hostname hq-srv.au-team.irpo
```

```
timedatectl set-timezone Asia/Yekaterinburg
```

```
mcedit /etc/net/ifaces/ens18/options → 's/DISABLED=yes/DISABLED=no'
```

```
echo "192.168.1.2/27" > /etc/net/ifaces/ens18/ipv4address
```

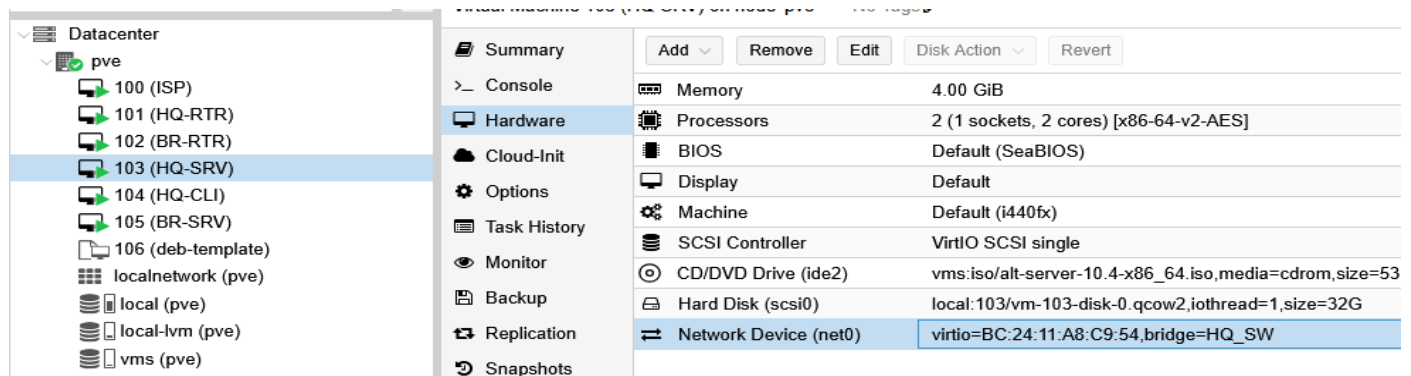
```
echo "default via 192.168.1.1" > /etc/net/ifaces/ens18/ipv4route
```

```
echo "nameserver 8.8.8.8" >> /etc/resolv.conf
```

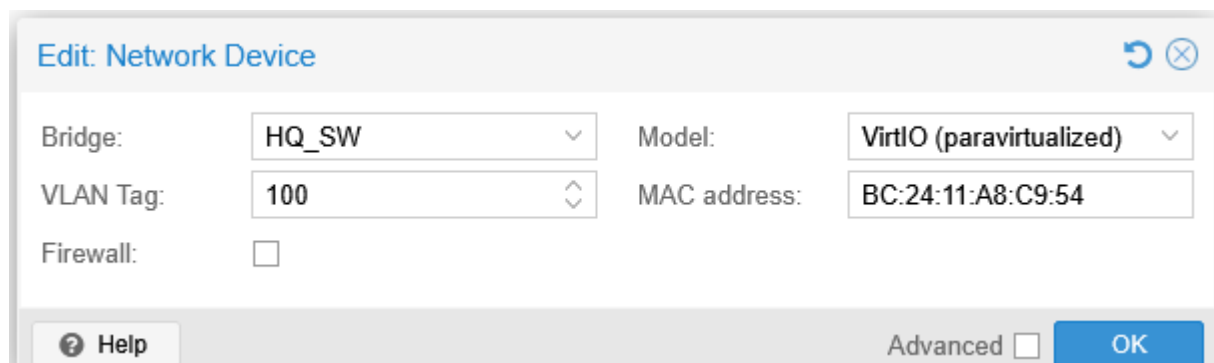
```
systemctl restart network
```

далее добавим vlan-тег к интерфейсу proxmox

дабл клик по интерфейсу машины



ставим VLAN Tag: 100



```
useradd sshuser -u 2026
```

```
passwd sshuser
```

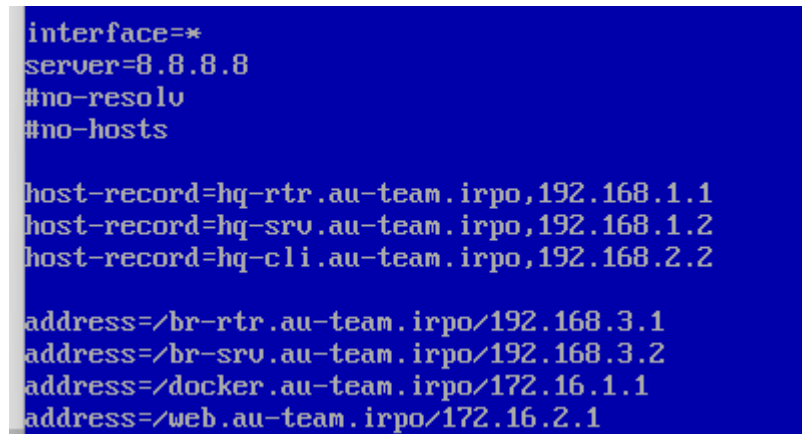
```
echo "sshuser ALL=(ALL) NOPASSWD:ALL" > /etc/sudoers.d/sshuser
```

```
chmod 4755 /usr/bin/sudo
```

DNS

```
apt-get update && apt-get install dnsmasq -y
```

```
mcedit /etc/dnsmasq.conf
```

A screenshot of a terminal window showing the configuration of the dnsmasq file. The text is as follows:

```
interface=*
server=8.8.8.8
#no-resolv
#no-hosts

host-record=hq-rtr.au-team.irpo,192.168.1.1
host-record=hq-srv.au-team.irpo,192.168.1.2
host-record=hq-cli.au-team.irpo,192.168.2.2

address=/br-rtr.au-team.irpo/192.168.3.1
address=/br-srv.au-team.irpo/192.168.3.2
address=/docker.au-team.irpo/172.16.1.1
address=/web.au-team.irpo/172.16.2.1
```

```
Systemctl restart dnsmasq
```

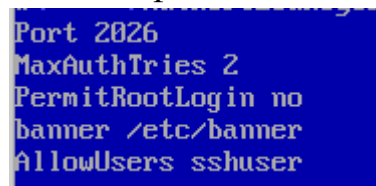
```
Systemctl enable --now dnsmasq
```

SSH

```
apt-get install openssh-server -y
```

```
mcedit /etc/openssh/sshd_config
```

В конце файла пишем:

A screenshot of a terminal window showing the configuration of the sshd_config file. The text is as follows:

```
Port 2026
MaxAuthTries 2
PermitRootLogin no
banner /etc/banner
AllowUsers sshuser
```

```
echo "Authorized access only" > /etc/banner
```

```
systemctl restart sshd
```

```
EDITOR=mcedit crontab -e
```

```
# в конец добавить:
```

```
@reboot /bin/systemctl restart network
```

BR-SRV(

Login: root

Password: P@ssw0rd

):

hostnamectl set-hostname br-srv.au-team.irpo

timedatectl set-timezone Asia/Yekaterinburg

mcedit /etc/net/iface/ens18/options → 's/DISABLED=yes/DISABLED=**no**/'

echo "192.168.3.2/27" > /etc/net/iface/ens18/ipv4address

echo "default via 192.168.3.1" > /etc/net/iface/ens18/ipv4route

echo "nameserver 8.8.8.8" >> /etc/resolv.conf

systemctl restart network

useradd sshuser -u 2026

passwd sshuser

echo "sshuser ALL=(ALL) NOPASSWD:ALL" > /etc/sudoers.d/sshuser

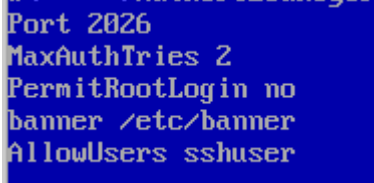
chmod 4755 /usr/bin/sudo

SSH

Apt-get update && apt-get install openssh-server -y

mcedit /etc/openssh/sshd_config

В конце файла пишем:



```
Port 2026
MaxAuthTries 2
PermitRootLogin no
banner /etc/banner
AllowUsers sshuser
```

echo "Authorized access only" > /etc/banner

systemctl restart sshd

```
EDITOR=mcedit crontab -e
```

[# в конец добавить:](#)

```
@reboot /bin/systemctl restart network
```

HQ-CLI(

Login: user

Password: P@ssw0rd

):

Открываем терминал.

Su

Вводим пароль: P@ssw0rd

hostnamectl set-hostname hq-cli.au-team.irpo

timedatectl set-timezone Asia/Yekaterinburg

ip a

Модуль 2. Настройка сетевой инфраструктуры

ISP (

Login: user

Password: P@ssw0rd

):

Sudo su

Chrony

apt-get update && apt-get install chrony -y

mcedit /etc/chrony/chrony.conf

Комментируем старый ntp:

```
# Welcome to the chrony configuration file. See chrony.conf(5) for more
# information about usable directives.
#pool 2.debian.pool.ntp.org iburst
```

И в конец файла пишем:

```
server 0.ru.pool.ntp.org iburst prefer minstratum 4
local stratum 5
allow 0.0.0.0/0
```

systemctl restart chronyd

apt-get install ssh nginx apache2-utils -y

htpasswd -bc /etc/nginx/.htpasswd WEB 'P@ssw0rd'

chmod 644 /etc/nginx/.htpasswd

rm -rf /etc/nginx/sites-available/default

mcedit /etc/nginx/sites-available/default

```

/etc/nginx/sites-available/default [----] 0 L:[ 1+27 28/ 28] *(669 / 669b) <EOF>
server {
<----->listen 80;
<----->server_name web.au-team.irpo;
<----->
<----->location / {
<-----><----->proxy_pass http://172.16.1.2:8080;
<-----><----->proxy_set_header Host $host;
<-----><----->proxy_set_header X-Real-IP $remote_addr;
<-----><----->proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
<-----><----->proxy_set_header X-Forwarded-Proto $scheme;
<-----><----->auth_basic 'Web-Authorization';
<-----><----->auth_basic_user_file /etc/nginx/.htpasswd;
<----->}
}
server {
<----->listen 80;
<----->server_name docker.au-team.irpo;
<----->
<----->location / {
<-----><----->proxy_pass http://172.16.2.2:8080;
<-----><----->proxy_set_header Host $host;
<-----><----->proxy_set_header X-Real-IP $remote_addr;
<-----><----->proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
<-----><----->proxy_set_header X-Forwarded-Proto $scheme;
<----->}
}

```

или

cd /etc/nginx/sites-available/

wget <http://file.danildebil.ru:1234/default>

nginx -t

systemctl restart nginx

HQ-RTR (

Login: user

Password: P@ssw0rd):

Sudo su

Apt-get update && apt-get install ssh

```
iptables -t nat -A PREROUTING -d 172.16.1.2 -p tcp --dport 8080 -j DNAT --to-destination 192.168.1.2:80
```

```
iptables -t nat -A PREROUTING -d 172.16.1.2 -p tcp --dport 2026 -j DNAT --to-destination 192.168.1.2:2026
```

```
iptables-save > /etc/iptables.rules
```

BR-RTR: (

Login: user

Password: P@ssw0rd):

Sudo su

Chrony

apt-get update && apt-get install ssh chrony -y

timedatectl set-ntp false

mcedit /etc/chrony/chrony.conf

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf
# Welcome to the chrony configuration file. See chrony.conf(5) for more
# information about usable directives.
#pool 2.debian.pool.ntp.org iburst ←
# This directive specify the location of the file containing ID/key pairs for
# NTP authentication.
keyfile /etc/chrony/chrony.keys
# This directive specify the file into which chronyd will store the rate
# information.
driftfile /var/lib/chrony/chrony.drift
# Uncomment the following line to turn logging on.
#log tracking measurements statistics
# Log files location.
logdir /var/log/chrony
# Stop bad estimates upsetting machine clock.
maxupdateskew 100.0
# This directive enables kernel synchronisation (every 11 minutes) of the
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.
#rtcsync ←
```

В конце прописываем:

```
server 172.16.1.1 iburst
```

Сохраняем chrony.conf

Перезагружаем и добавляем в автозагрузку chrony:

systemctl restart chrony

iptables -t nat -A PREROUTING -d 172.16.2.2 -p tcp --dport 8080 -j DNAT --to-destination 192.168.3.2:8080

```
iptables -t nat -A PREROUTING -d 172.16.2.2 -p tcp --dport 2026 -j DNAT --to-destination 192.168.3.2:2026  
iptables-save > /etc/iptables.rules
```

HQ-SRV(

Login: root

Password: P@ssw0rd

):

DNS:

mcedit /etc/dnsmasq.conf

Добавите строку над server = 8.8.8.8:

```
server=/au-team.irpo/192.168.3.2  
server=8.8.8.8
```

Chrony

apt-get update && apt-get install chrony -y

timedatectl set-ntp false

mcedit /etc/chrony.conf

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf  
  
# Welcome to the chrony configuration file. See chrony.conf(5) for more  
# information about usable directives.  
#pool 2.debian.pool.ntp.org iburst  
  
# This directive specify the location of the file containing ID/key pairs for  
# NTP authentication.  
keyfile /etc/chrony/chrony.keys  
  
# This directive specify the file into which chronyd will store the rate  
# information.  
driftfile /var/lib/chrony/chrony.drift  
  
# Uncomment the following line to turn logging on.  
#log tracking measurements statistics  
  
# Log files location.  
logdir /var/log/chrony  
  
# Stop bad estimates upsetting machine clock.  
maxupdateskew 100.0  
  
# This directive enables kernel synchronisation (every 11 minutes) of the  
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.  
#rtcsync
```

В конце прописываем:

```
server 172.16.1.1 iburst
```

Сохраняем chrony.conf

Перезагружаем и добавляем в автозагрузку chrony:
systemctl restart chronyd

RAID

```
mdadm --create /dev/md0 --level=0 --raid-device=2 /dev/sdb /dev/sdc --run  
fdisk /dev/md0
```

n

p

enter

enter

w

```
mkfs.ext4 /dev/md0p1  
mdadm --detail --scan >> /etc/mdadm.conf
```

```
mkdir /raid
```

```
echo "/dev/md0p1 /raid ext4 defaults 0 2" >> /etc/fstab
```

```
mount -a
```

NFS

```
apt-get install nfs-server -y
```

```
mkdir -p /raid/nfs  
chmod 777 /raid/nfs  
echo "/raid/nfs 192.168.2.0/28(rw,sync,no_subtree_check)" > /etc/exports  
exportfs -ra
```

```
systemctl restart nfs-server  
systemctl enable --now nfs-server
```

LAMP

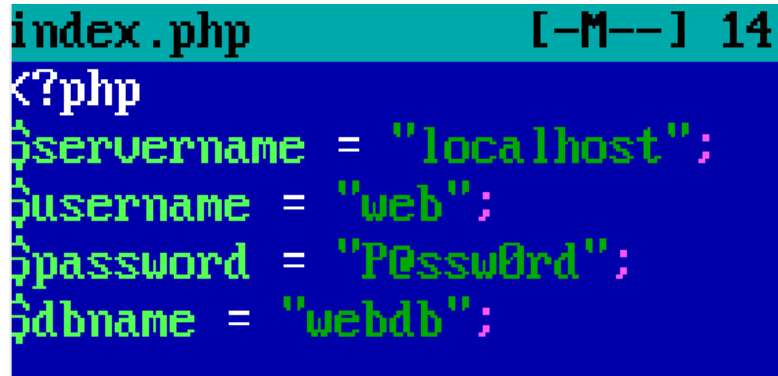
```
apt-get install lamp-server -y
```

```
mount /dev/sr0 /mnt/
```

```
cp /mnt/web/index.php /var/www/html  
cp /mnt/web/logo.png /var/www/html
```

```
chown -R apache:apache /var/www/html
chmod -R 755 /var/www/html
```

```
mcedit /var/www/html/index.php
```

A screenshot of a text editor window. The title bar at the top is teal and contains the text 'index.php' followed by a window control icon and the text '[-M--] 14'. The editor area has a dark blue background. The text inside is written in a monospaced font with syntax highlighting: the opening PHP tag '<?php' is in white, and the database connection variables are in green. The code is: '\$servername = "localhost";', '\$username = "web";', '\$password = "P@ssw0rd";', and '\$dbname = "webdb";'.

```
index.php [-M--] 14
<?php
$servername = "localhost";
$username = "web";
$password = "P@ssw0rd";
$dbname = "webdb";
```

```
systemctl enable --now mariadb
mariadb -u root
```

```
CREATE DATABASE webdb;
CREATE USER 'web'@'localhost' IDENTIFIED BY 'P@ssw0rd';
GRANT ALL PRIVILEGES ON webdb.* TO 'web'@'localhost' WITH GRANT
OPTION;
FLUSH PRIVILEGES;
Exit
```

```
mariadb -u root webdb < /mnt/web/dump.sql
```

```
systemctl restart httpd2
systemctl enable --now httpd2
```

BR-SRV(

Login: root

Password: P@ssw0rd

):

Chrony

timedatectl set-ntp false

mcedit /etc/chrony.conf

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf
# Welcome to the chrony configuration file. See chrony.conf(5) for more
# information about usable directives.
#pool 2.debian.pool.ntp.org iburst
# This directive specify the location of the file containing ID/key pairs for
# NTP authentication.
keyfile /etc/chrony/chrony.keys
# This directive specify the file into which chronyd will store the rate
# information.
driftfile /var/lib/chrony/chrony.drift
# Uncomment the following line to turn logging on.
#log tracking measurements statistics
# Log files location.
logdir /var/log/chrony
# Stop bad estimates upsetting machine clock.
maxupdateskew 100.0
# This directive enables kernel synchronisation (every 11 minutes) of the
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.
#rtcsync
```

В конце прописываем:

```
server 172.16.1.1 iburst
```

Сохраняем chrony.conf

Перезагружаем и добавляем в автозагрузку chrony:

systemctl restart chronyd

SAMBA DC

apt-get update && apt-get install task-samba-dc -y

rm -f /etc/samba/smb.conf

samba-tool domain provision

cp /var/lib/samba/private/krb5.conf /etc/krb5.conf

mcedit /etc/samba/smb.conf

Добавить:



```
----->netbios name = BR-SRU
----->realm = AU-TEAM.IRPO
----->server role = active directory domain controller
----->workgroup = AU-TEAM
----->interfaces = lo ens18
```

systemctl restart samba (может быть долгим, мб ctrl+c и заново)

systemctl enable --now samba

samba-tool user create hquser1 P@ssw0rd

samba-tool user create hquser2 P@ssw0rd

samba-tool user create hquser3 P@ssw0rd

samba-tool user create hquser4 P@ssw0rd

samba-tool user create hquser5 P@ssw0rd

samba-tool group add hq

samba-tool group addmembers hq hquser1

samba-tool group addmembers hq hquser2

samba-tool group addmembers hq hquser3

samba-tool group addmembers hq hquser4

samba-tool group addmembers hq hquser5

Docker

apt-get install docker-compose docker-engine -y

systemctl enable --now docker

mount /dev/sr0 /mnt/

mkdir -p /root/docker/

cp -r /mnt/docker /root/

docker image load -i /root/docker/site_latest.tar

docker image load -i /root/docker/mariadb_latest.tar

mkdir -p /root/testapp/

mcedit /root/testapp/docker-compose.yaml

```

docker-compose.yml [-M--] 0 L:[ 1+26 27/ 27] *(611 / 611b) <EOF>
services:
  testapp:
    image: site:latest
    container_name: testapp
    restart: always
    depends_on:
      - db
    ports:
      - 8080:8080
    environment:
      DB_TYPE: maria
      DB_HOST: db
      DB_NAME: testdb
      DB_PORT: 3306
      DB_USER: test
      DB_PASS: P@ssw0rd
  .....
  db:
    image: mariadb:10.11
    container_name: db
    restart: always
    environment:
      MARIADB_DATABASE: testdb
      MARIADB_USER: test
      MARIADB_PASSWORD: P@ssw0rd
      MARIADB_ROOT_PASSWORD: P@ssw0rd

```

Или

Apt-get install wget -y

Cd /root/testapp/

wget <http://file.danildebil.ru:1234/docker-compose.yml>

docker compose up -d

МБ надо в cron добавить:

crontab -e

```

crontab -e
docker restart db
docker restart testapp

```

Ansible

apt-get install ansible sshpass -y

mcedit /etc/ansible/hosts

```

hosts [-M--] 45 L:[ 1+ 4 5/ 5] *(346 / 392b) 0117 0x075
[clients]
hq-rtr ansible_host=192.168.1.1 ansible_user=net_admin ansible_port=22 ansible_password=P@ssw0rd
br-rtr ansible_host=192.168.3.1 ansible_user=net_admin ansible_port=22 ansible_password=P@ssw0rd
hq-srv ansible_host=192.168.1.2 ansible_user=sshuser ansible_port=2026 ansible_password=P@ssw0rd
hq-cli ansible_host=192.168.2.2 ansible_user=user ansible_port=22 ansible_password=P@ssw0rd

```

rm -rf /etc/ansible/ansible.cfg

mcedit /etc/ansible/ansible.cfg

```
ansible.cfg      [-M--] 30 L:[ 1+ 3  4/  4] *(96 / 96b) <EOF>
[defaults]
interpreter_python = python3
host_key_checking = False
inventory = /etc/ansible/hosts
```

HQ-CLI(

Login: user

Password: P@ssw0rd

):

Открываем терминал.

Su

Вводим пароль: P@ssw0rd

Chrony

apt-get update

timedatectl set-ntp false

mcedit /etc/chrony.conf

Комментируем следующие строки:

```
GNU nano 2.7.4                                Файл: /etc/chrony/chrony.conf
# Welcome to the chrony configuration file. See chrony.conf(5) for more
# information about usable directives.
#pool 2.debian.pool.ntp.org iburst ←
# This directive specify the location of the file containing ID/key pairs for
# NTP authentication.
keyfile /etc/chrony/chrony.keys
# This directive specify the file into which chronyd will store the rate
# information.
driftfile /var/lib/chrony/chrony.drift
# Uncomment the following line to turn logging on.
#log tracking measurements statistics
# Log files location.
logdir /var/log/chrony
# Stop bad estimates upsetting machine clock.
maxupdateskew 100.0
# This directive enables kernel synchronisation (every 11 minutes) of the
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.
#rtcsync ←
```

В конце прописываем:

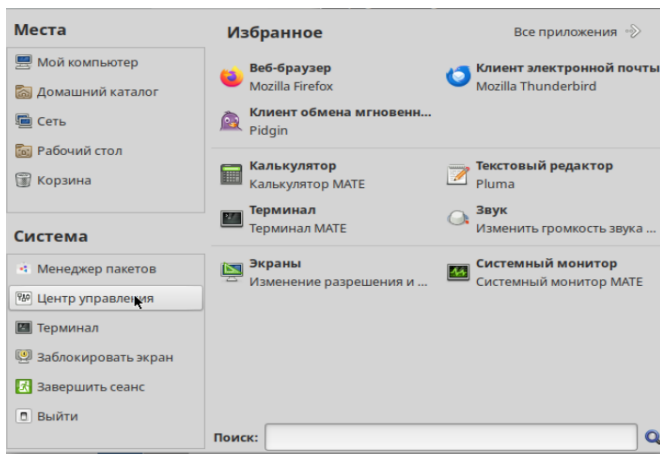
```
server 172.16.1.1 iburst
```

Сохраняем chrony.conf

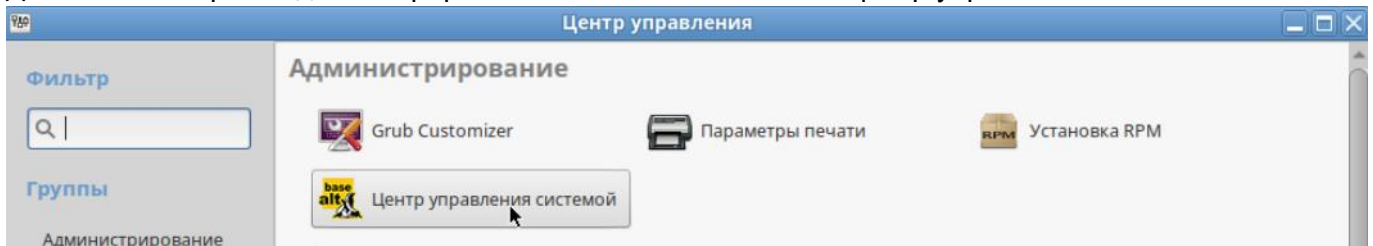
Перезагружаем и добавляем в автозагрузку chrony:

systemctl restart chronyd

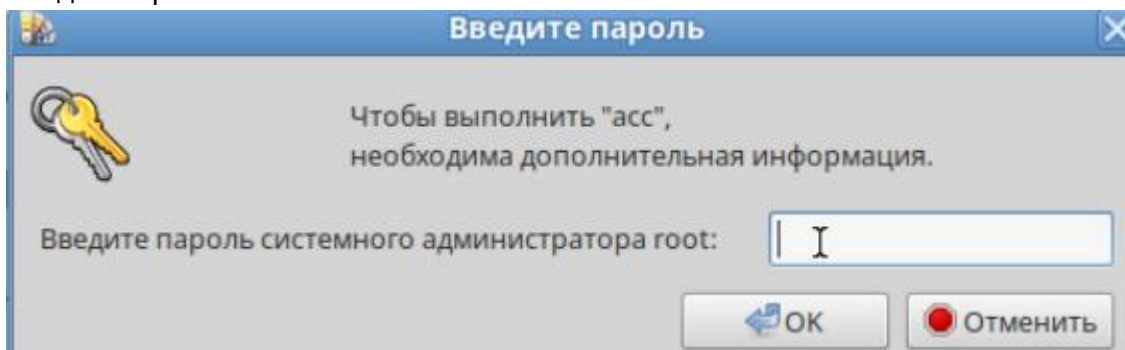
Ввод в домен



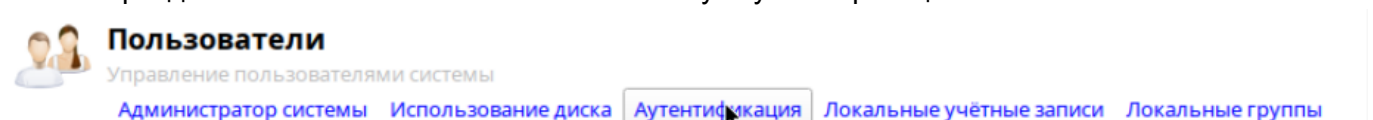
Далее в категории “Администрирование” кликаем по иконке “Центр управления системой”



Вводим пароль от root-пользователя

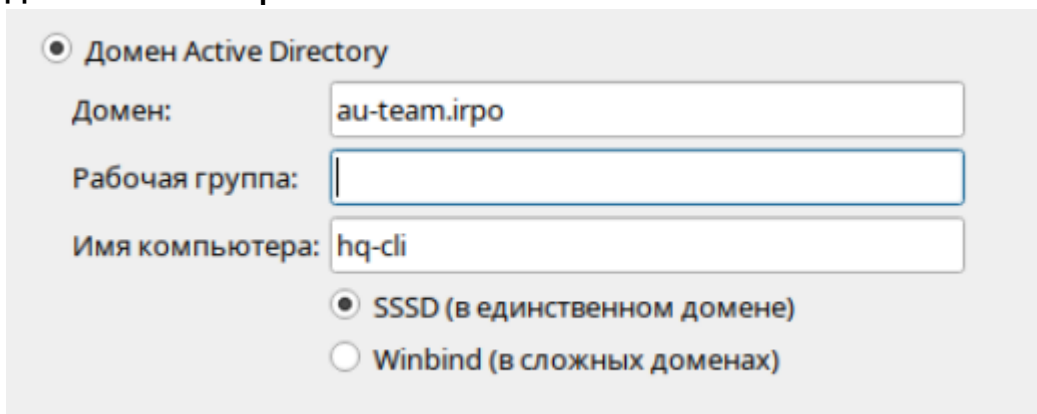


Затем в разделе “Пользователи” кликаем на иконку “Аутентификация”

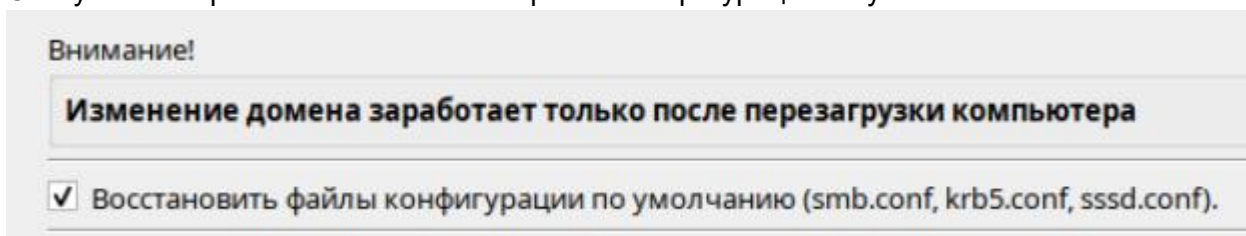


После выбираем пункт “Домен Active Directory” и заполняем поле домена

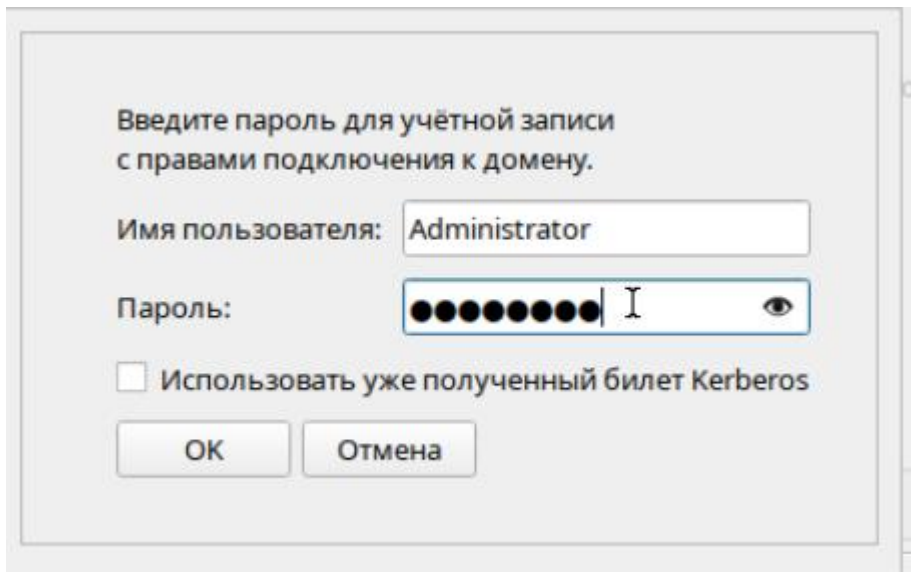
Домен: au-team.irpo



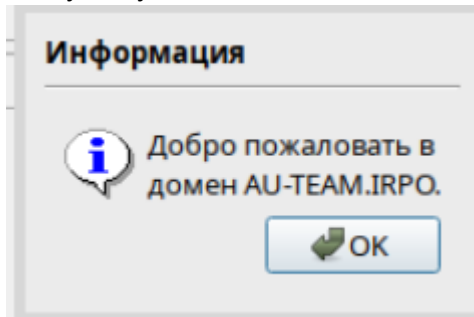
Снизу ставим флажок “Восстановить файлы конфигурации по умолчанию”



Далее в появившемся окне вводим созданный при развертывании контроллера домена пароль и нажимаем "ОК"



В случае успеха появится вот-такое окно!



Добавление прав hquser-ам

```
echo "%hq ALL=(ALL) NOPASSWD: /usr/bin/cat, /bin/grep, /usr/bin/id" >> /etc/sudoers.d/hq
```

NFS client

```
apt-get install nfs-clients -y
```

```
mkdir /mnt/nfs
```

```
echo "HQ-SRV:/raid/nfs /mnt/nfs nfs defaults,_netdev 0 0" >> /etc/fstab
```

↑ тут нижнее подчеркивание

```
mount -a
```

SSH

```
apt-get install openssh-server -y  
systemctl restart sshd
```

Browser

```
apt-get install yandex-browser-stable -y
```

ПРОВЕРКА

BR-SRV:

ansible all -m ping

HQ-CLI:

Заходит на <http://web.au-team.irpo>

Заходит на <http://docker.au-team.irpo>

Mkdir /mnt/nfs/asd

Hq-srv:

Ls /raid/nfs <- появилась, значит ГУУУУД

Модуль 3. Эксплуатация объектов сетевой инфраструктуры

Задание 1. Выполните импорт пользователей в домен au-team.irpo

BR-SRV:

```
mount /dev/sr0/ mnt/
{
apt-get update && apt-get install git
git clone https://github.com/Rogotochka/ModulearExam
cd ModulearExam
chmod +x Module3--EX1.sh
./Module3--EX1.sh
Ждём
}
```

Или

```
{
apt-get update && apt-get install wget
wget http://file.danildebil.ru:1234/ModulearExam/Module3--EX1.sh
chmod +x Module3--EX1.sh
./Module3--EX1.sh
Ждём
}
```

Задание 2. Выполните настройку центра сертификации на базе HQ-SRV

ISP:

```
apt-get install openssh-server  
systemctl restart sshd
```

HQ-SRV:

```
{  
apt-get update && apt-get install git  
git clone https://github.com/Rogotochka/ModulearExam  
cd ModulearExam  
chmod +x certificate.sh  
./certificate.sh
```

Может быть ошибка по одному из последних ssh подключений

```
}
```

ISP:

```
cp /home/user/{web,docker}.au-team.irpo.{key,cer} /etc/nginx/ssl/
```

```
rm -rf /etc/nginx/sites-available/default
```

```
{  
apt-get update && apt-get install git  
git clone https://github.com/Rogotochka/ModulearExam  
cd ModulearExam  
chmod +x Module3--EX2.sh  
./Module3--EX2.sh
```

```
}
```

ИЛИ

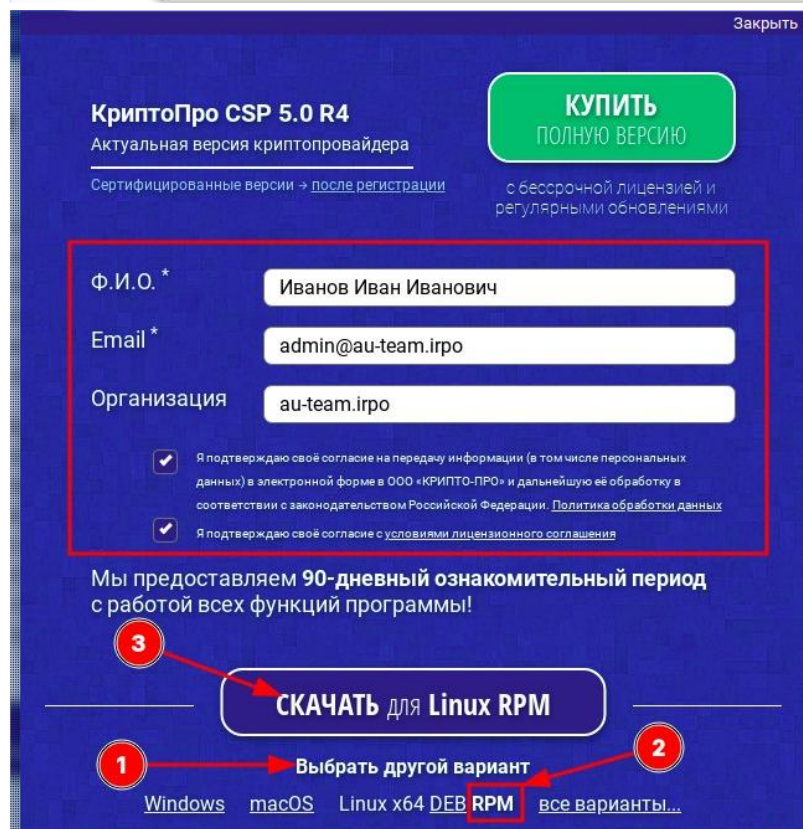
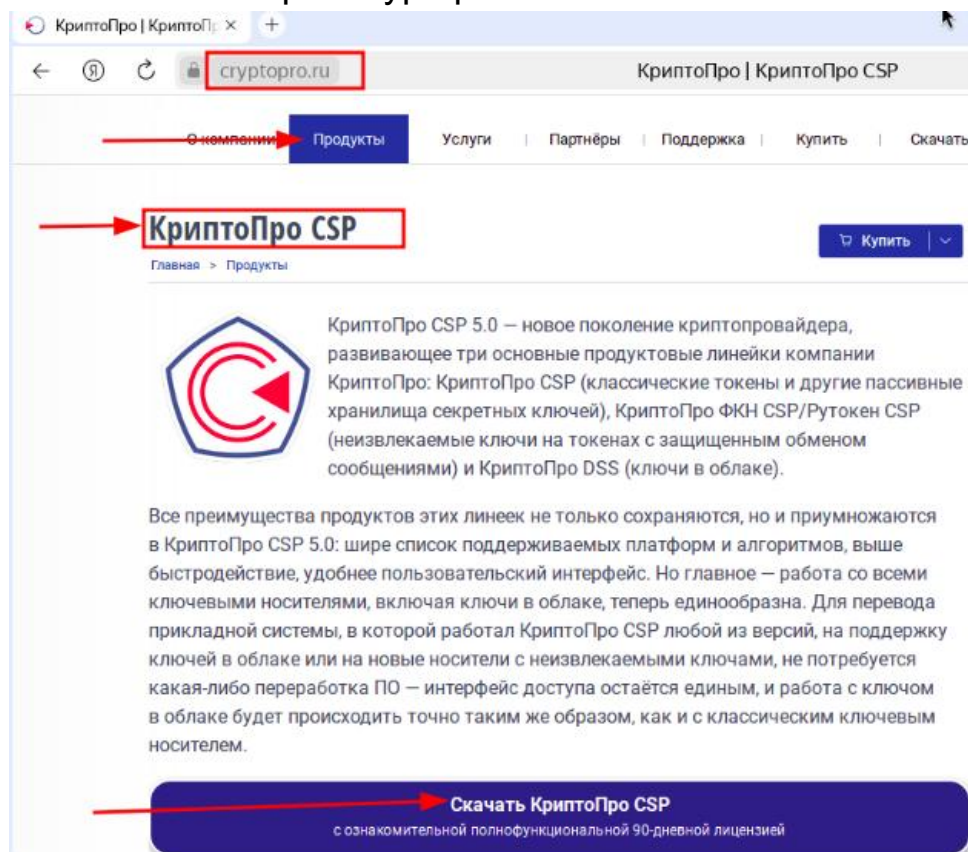
```
{  
wget http://file.danildebil.ru:1234/ModulearExam/default  
cp ./default /etc/nginx/sites-available/default  
}
```

nginx -t
apt-get install libgost-astra

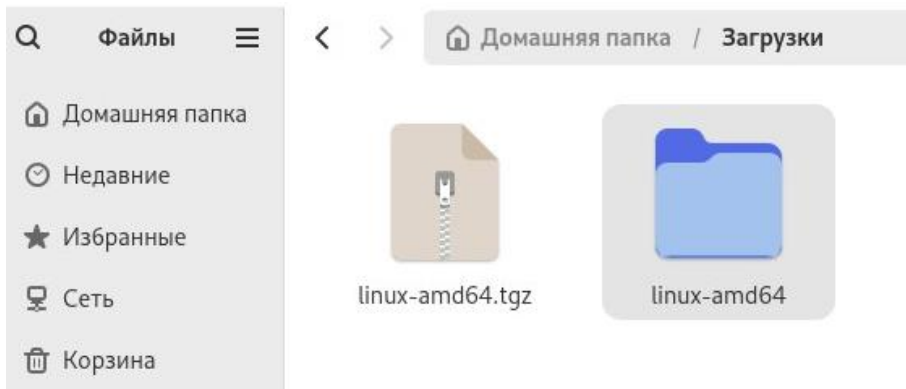
HQ-CLI:

cp /home/user/ca.cer /etc/pki/ca-trust/source/anchors/ && update-ca-trust

Скачиваем с <https://cryptopro.ru>



Распакуем архив

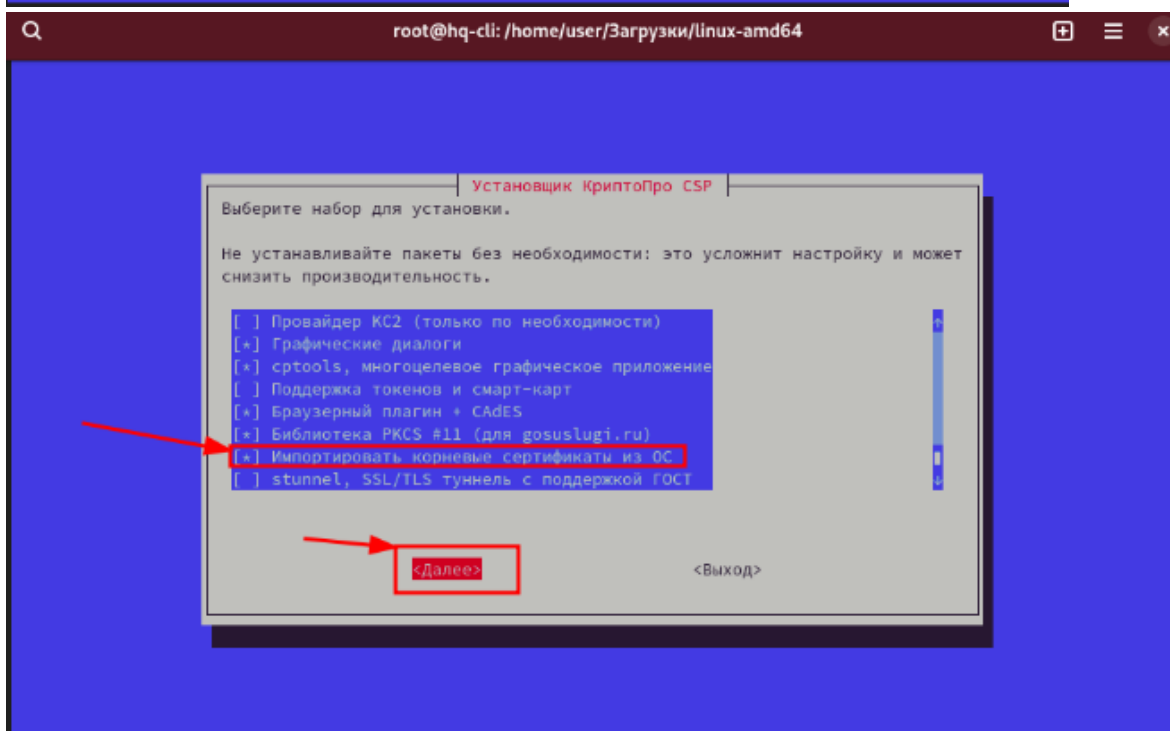
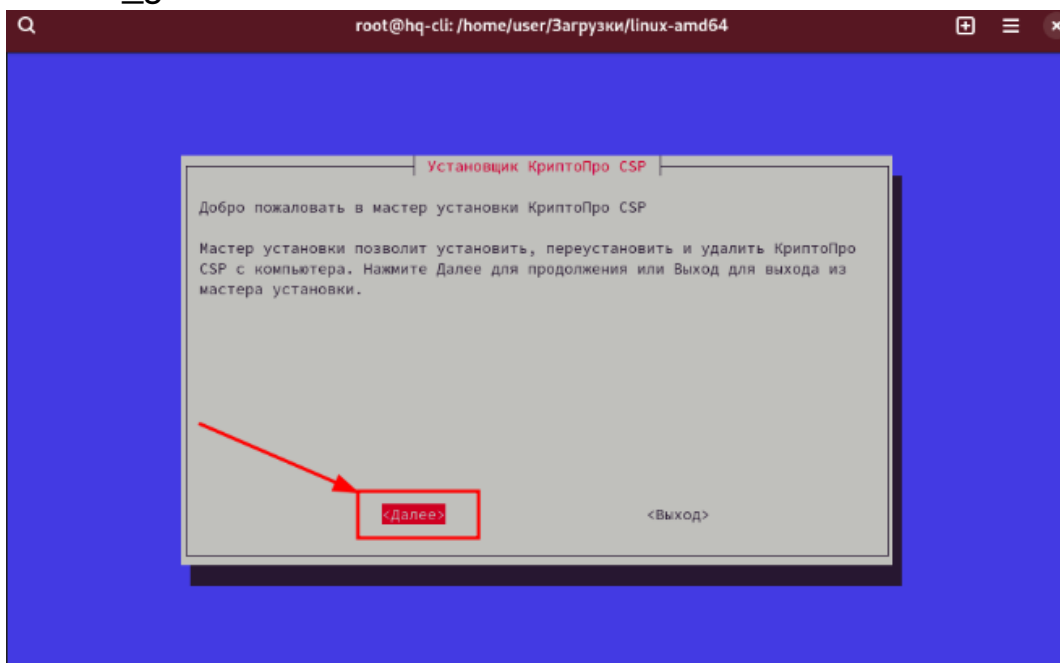


далее

```
cd /home/user/Загрузки/linux-amd64
```

```
apt-get install -y cryptopro-preinstall
```

```
./install_gui.sh
```



Там докликать
И проверить

Задание 3. Перенастройте ip-туннель с базового до уровня туннеля, обеспечивающего шифрование трафика

HQ-RTR:

```
Apt-get update && Apt-get install git  
Git clone https://github.com/Rogotochka/modulearExam  
Cd modulearExam  
Chmod +x HQ-RTR-Module3--EX3.sh  
./HQ-RTR-Module3--EX3.sh
```

BR-RTR:

```
Apt-get update && Apt-get install git  
Git clone https://github.com/Rogotochka/modulearExam  
Cd modulearExam  
Chmod +x BR-RTR-Module3--EX3.sh  
./BR-RTR-Module3--EX3.sh
```

Проверить

```
Ipsec statusall
```

Задание 4. Перенастройте ip-туннель с базового до уровня туннеля, обеспечивающего шифрование трафика.

HQ-RTR:

```
Chmod +x firewall.sh  
./firewall.sh
```

BR-RTR:

```
Chmod +x firewall.sh  
./firewall.sh
```

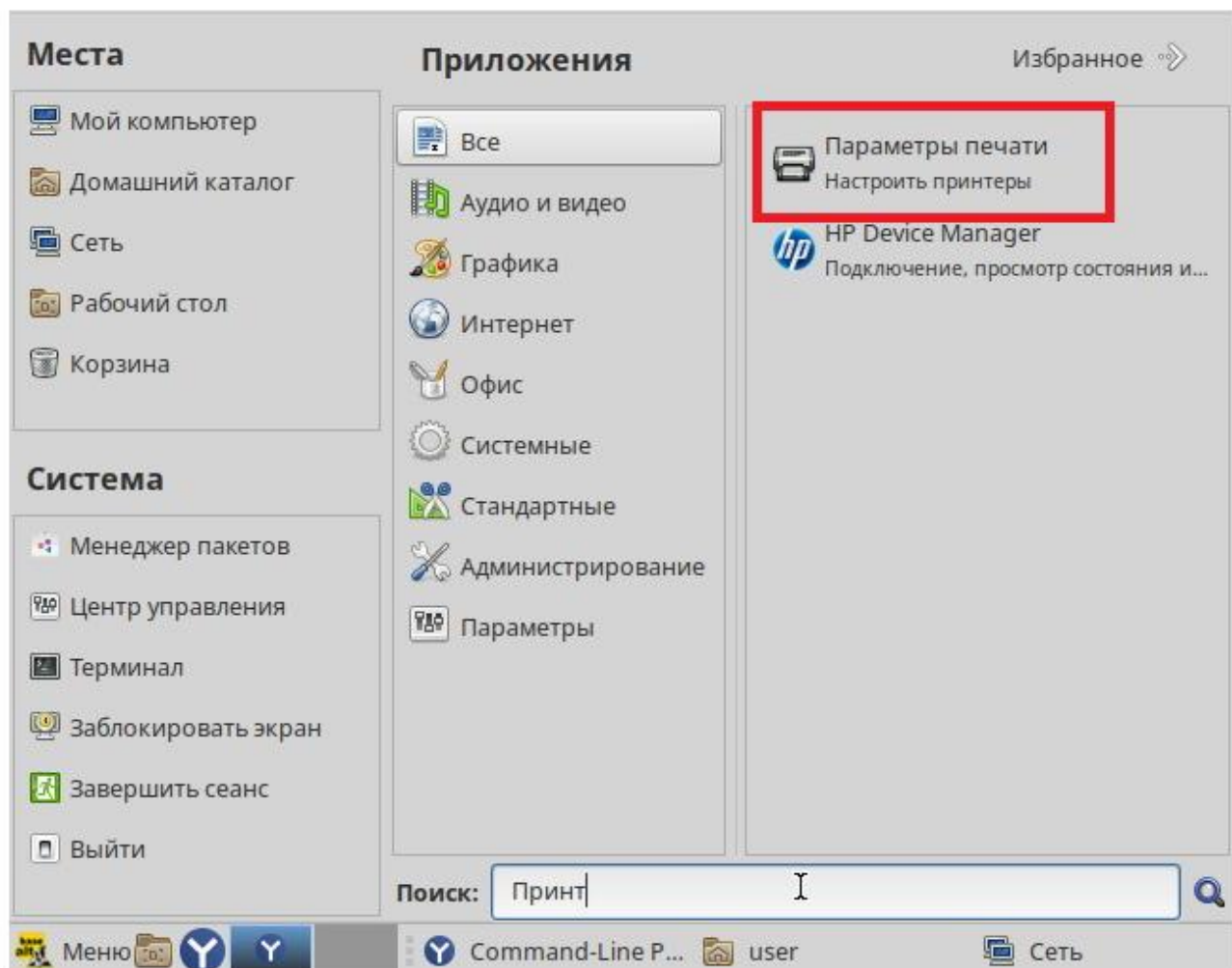
Задание 5. Настройте принт-сервер cups на сервере HQ-SRV:

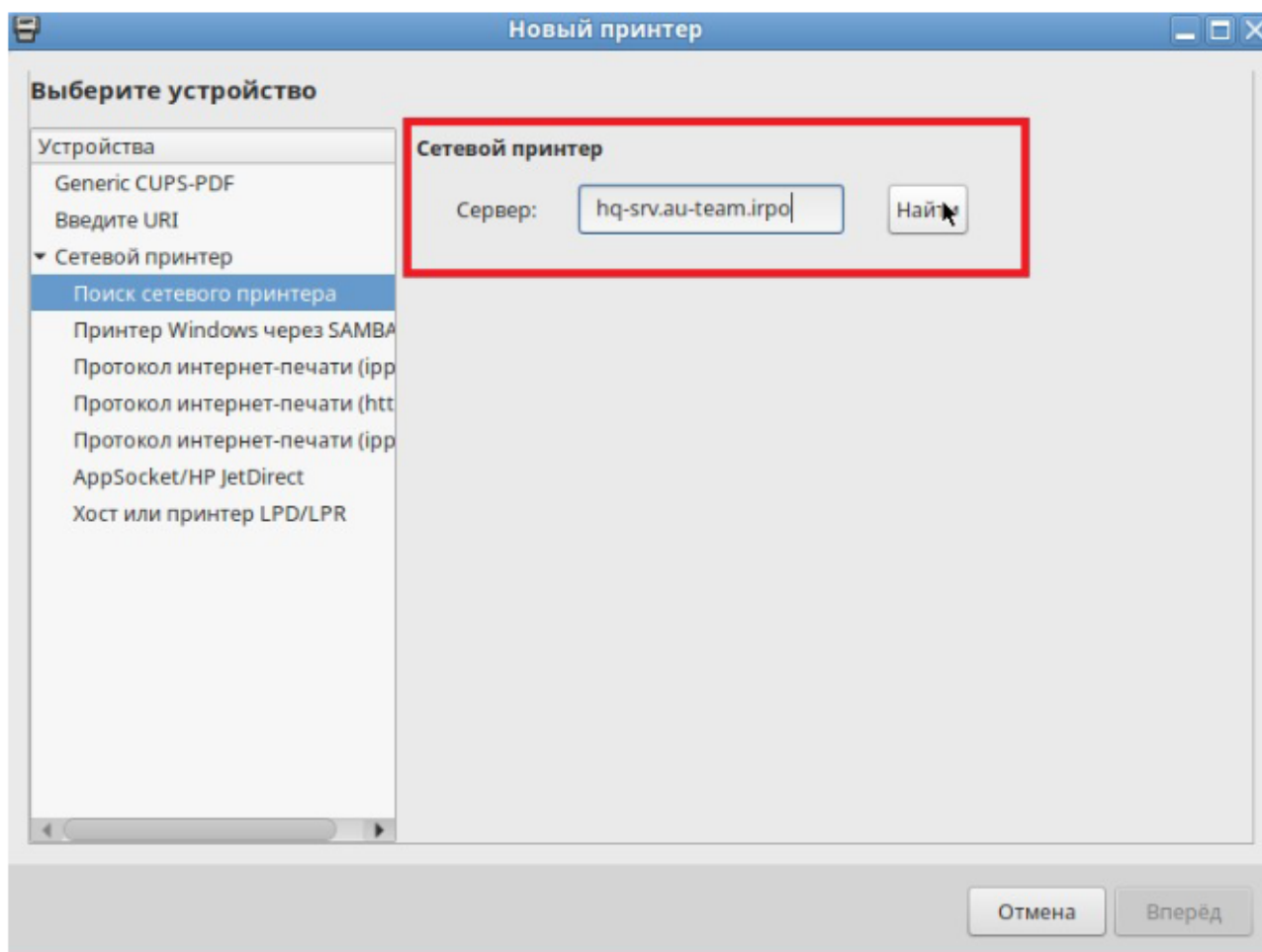
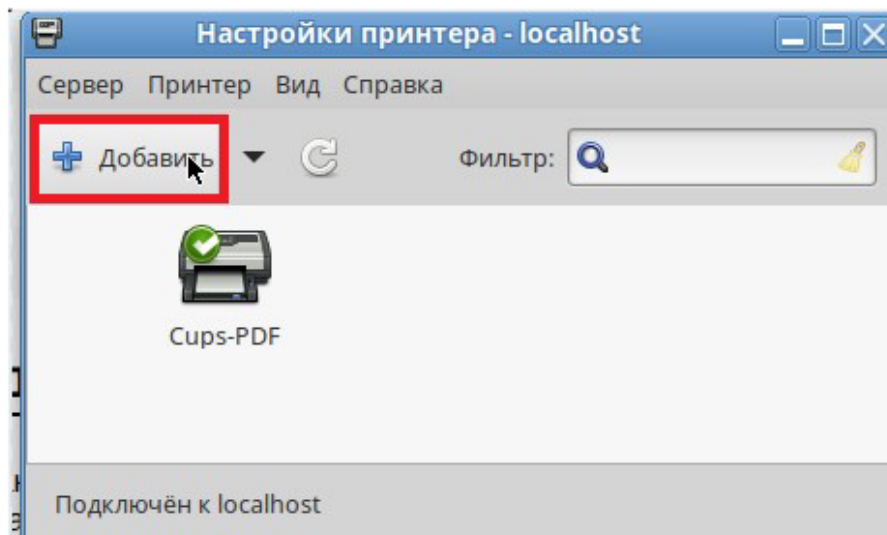
HQ-SRV:

```
apt-get update && apt-get install -y cups cups-pdf  
systemctl enable --now cups  
cupsctl --share-printers --remote-any
```

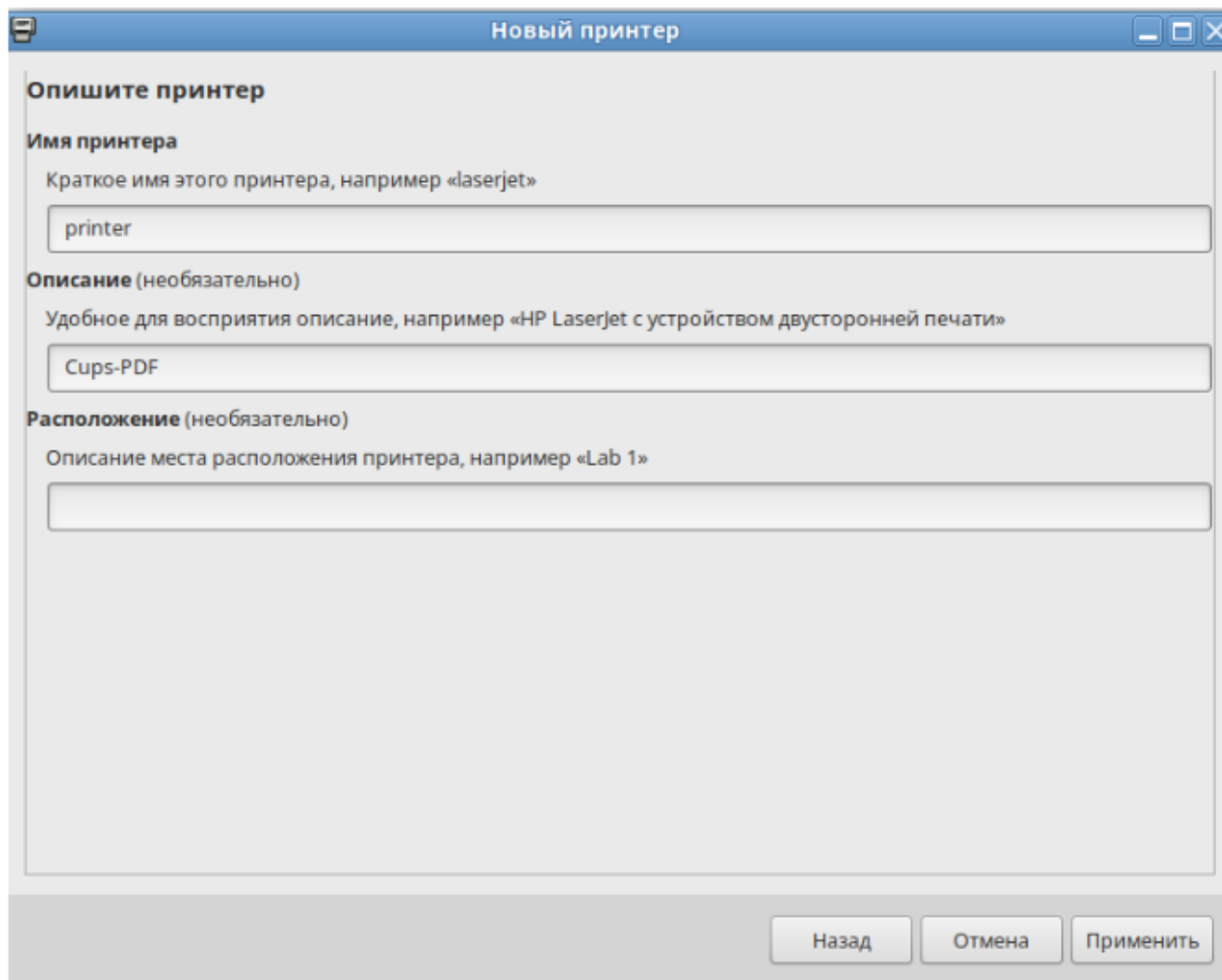
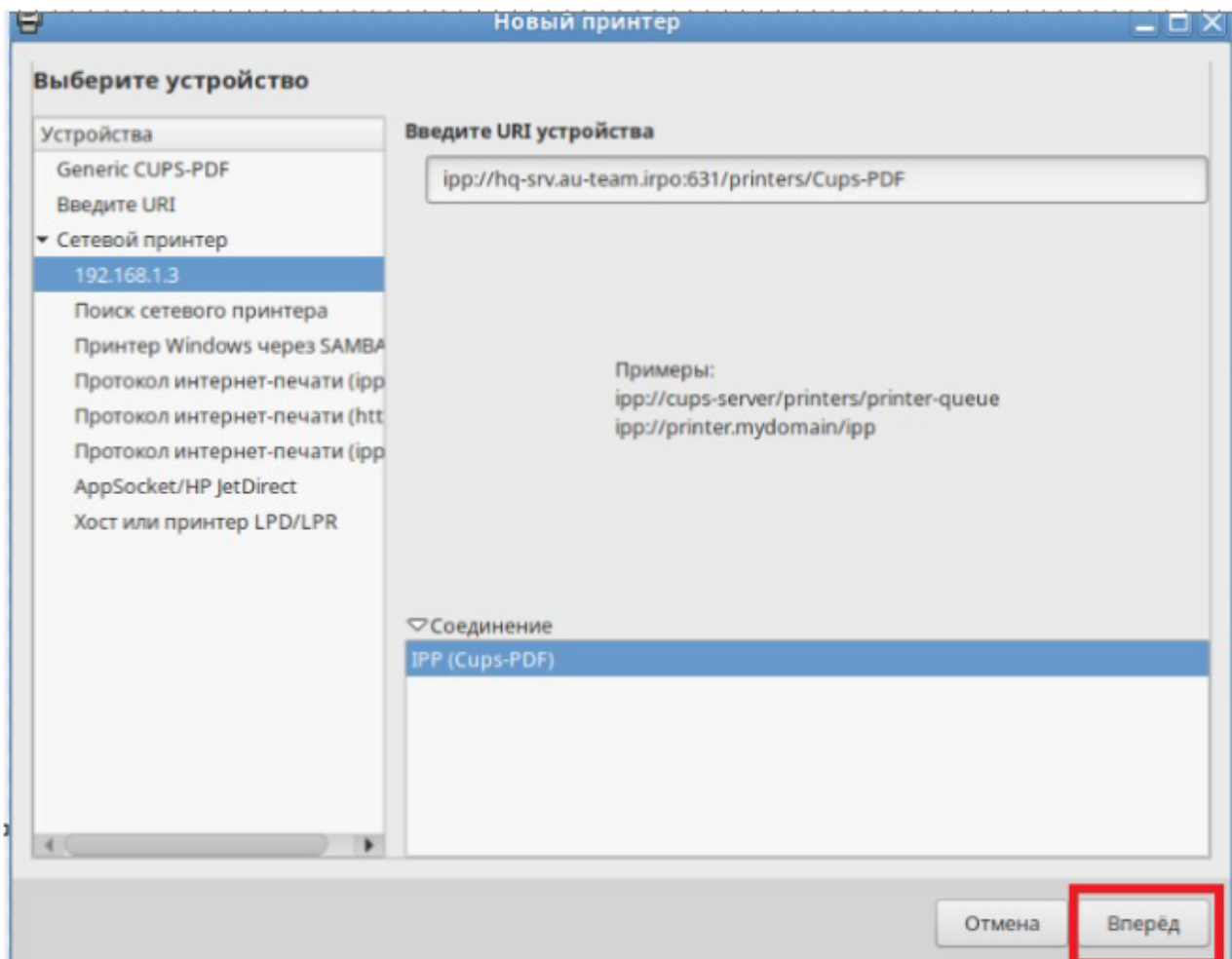
HQ-CLI:

Подключаем виртуальный принтер как принтер по умолчанию:





Если предложат авторизоваться на hq-srv, нажимаем отменить!



Новый принтер

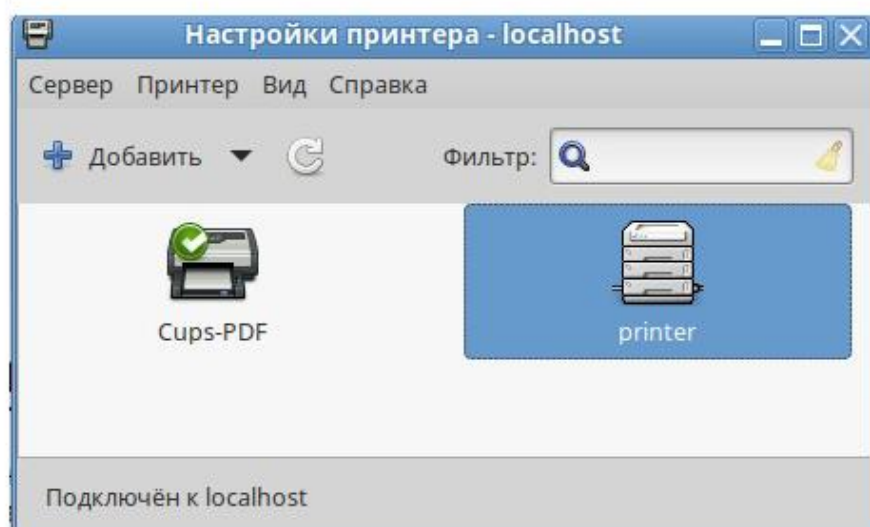
Опишите принтер

Имя принтера
Краткое имя этого принтера, например «laserjet»

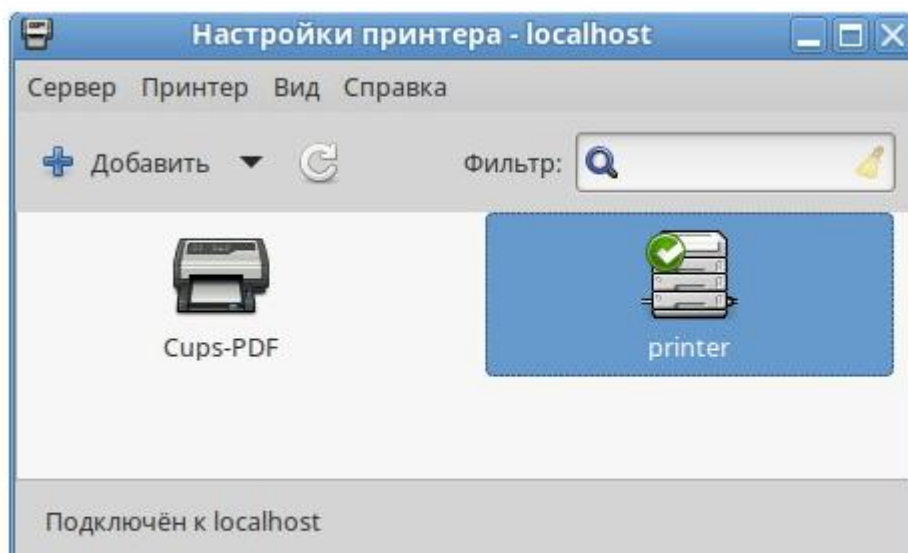
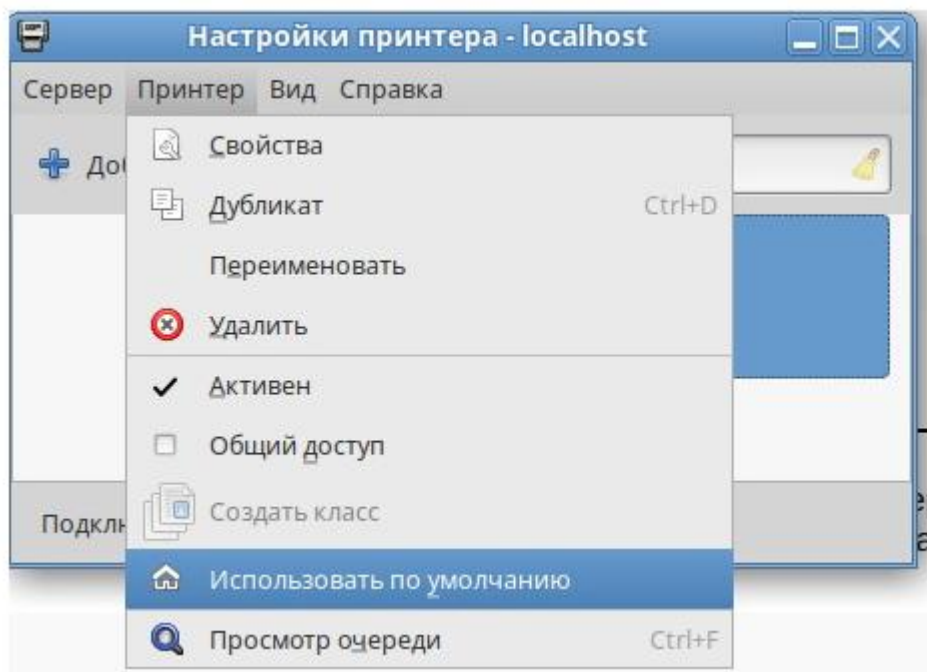
Описание (необязательно)
Удобное для восприятия описание, например «HP Laserjet с устройством двусторонней печати»

Расположение (необязательно)
Описание места расположения принтера, например «Lab 1»

Назад Отмена **Применить**



Далее необходимо сделать его принтером по умолчанию:



Проверяем печать:

Задание 6. Реализуйте логирование при помощи rsyslog на устройствах HQ-RTR, BR-RTR, BR-SRV

HQ-SRV:

```
Apt-get update && Apt-get install git  
Git clone https://github.com/Rogotochka/modulearExam  
Cd modulearExam  
Chmod +x Server-Module3--EX6.sh  
./Server-Module3--EX6.sh
```

BR-SRV:

```
Apt-get update && Apt-get install git  
Git clone https://github.com/Rogotochka/modulearExam  
Cd modulearExam  
Chmod +x Client-Module3--EX6.sh  
./Client-Module3--EX6.sh
```

BR-RTR:

```
Apt-get update && Apt-get install git  
Git clone https://github.com/Rogotochka/modulearExam  
Cd modulearExam  
Chmod +x Client-Module3--EX6.sh  
./Client-Module3--EX6.sh
```

HQ-RTR:

```
Apt-get update && Apt-get install git  
Git clone https://github.com/Rogotochka/modulearExam  
Cd modulearExam  
Chmod +x Client-Module3--EX6.sh  
./Client-Module3--EX6.sh
```

Задание 7. На сервере HQ-SRV реализуйте мониторинг устройств с помощью открытого программного обеспечения

BR-SRV:

Apt-get update && Apt-get install git

Git clone <https://github.com/Rogotochka/modulearExam>

Cd modulearExam

Chmod +x Client-Module3--EX7.sh

./Client-Module3--EX7.sh

HQ-SRV:

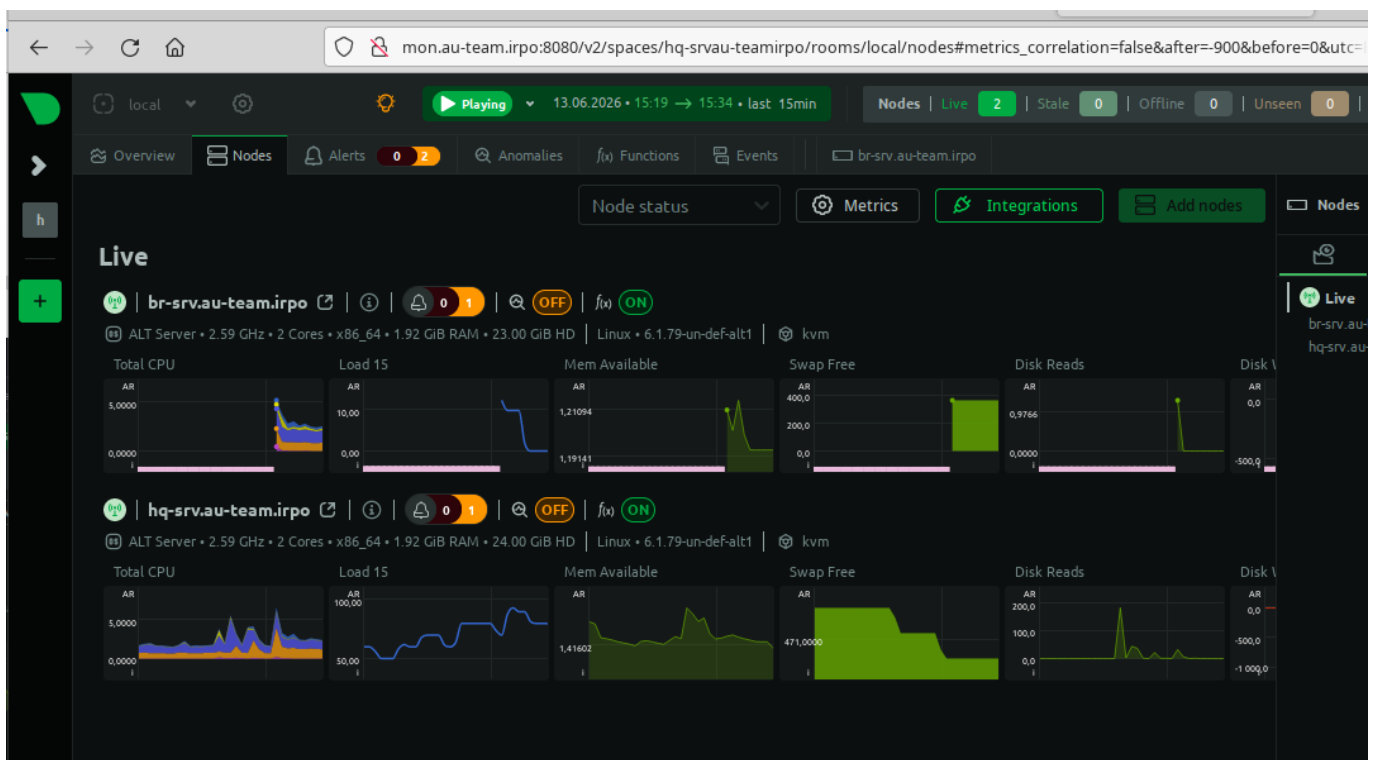
Apt-get update && Apt-get install git

Git clone <https://github.com/Rogotochka/modulearExam>

Cd modulearExam

Chmod +x Server-Module3--EX7.sh

./Server-Module3--EX7.sh



**Задание 8. Реализуйте механизм инвентаризации машин HQ-SRV и HQ-CLI через Ansible на BR-SRV:
BR-SRV:**

Apt-get update && Apt-get install git

Git clone <https://github.com/Rogotochka/modulearExam>

Cd modulearExam

Cp ./get_hostname_address.yml /etc/ansible

Mkdir /etc/ansible/PC-INFO

Ansible-playbook /etc/ansible/get_hostname_address.yml

Задание 9. На HQ-SRV настройте программное обеспечение fail2ban для защиты ssh

HQ-SRV:

Apt-get update && Apt-get install git

Git clone <https://github.com/Rogotochka/modulearExam>

Cd modulearExam

Chmod +x Module3--EX9.sh

./Module3--EX9.sh

Задание 10. Настройка резервного копирования директории сервера